

DSA Day 26: AVL deletion & Red-Black intro

Book: Data Structures Through C in Depth - Ch6 §6.14.4-6.15.1 · pp 248-260 | Code in Python

Overview

AVL deletion may need several rotations up the path. Red-black trees are an alternative balanced BST with color rules.

Key concepts to master

- Rebalance on delete
- 5 red-black properties
- Recolor vs rotate

Python implementation

```
# RB properties: root black; red node has black children;  
# every path root->leaf has equal black count; leaves(NIL) black
```

Complexity

$O(\log n)$.

Common pitfalls

- Stopping rebalancing too early on delete

Practice

- Delete from your AVL and re-check balance
- List the red-black properties

